

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

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Abstract

001 Segmentation models in medical imaging are trained on one
002 acquisition setting and must generalize to others. Label-
003 generative domain randomization, of which SynthSeg is the
004 archetype, addresses this by discarding the acquired scan
005 and resampling intensities from a per-label Gaussian mixture
006 over a dense anatomical label map – an annotation far
007 richer than the segmentation task itself provides, and one
008 that does not preserve texture: real gradients and partial-
009 volume detail become piecewise noise. We ask when that
010 loss costs accuracy, and find the answer is a property of the
011 task. Where the target is separated from its surroundings
012 by an anatomical interface – encapsulated abdominal or-
013 gans, bounded by fat planes – destroying texture costs nothing.
014 Where no interface exists – diffuse glioma, infiltrating
015 beyond both the enhancing margin and the peritumoral
016 FLAIR abnormality – the target must be inferred from tissue
017 appearance, and destroying texture is expensive. We
018 exploit this with PALETTE, a single-source augmentation
019 needing no dense label map: it partitions a scan’s fore-
020 ground by unsupervised k -means, sub-divides each class
021 spatially by a Voronoi partition, and applies a signed ran-
022 dom affine remap to the real intensities of each sub-region.
023 Remapping existing intensities rather than replacing them
024 leaves edges, gradients and partial-volume detail largely
025 intact; only the contrast relationships between regions are
026 randomized. A controlled ablation isolates this: holding
027 the partition and per-region statistics fixed and swapping
028 only the real-intensity remap for label-style noise fill costs
029 7.7 and 7.2 Dice on the two tasks with no tissue interface,
030 against +1.1 and –1.1 on the two with one. Across cross-
031 contrast and cross-modality transfer on healthy-brain, MS-
032 lesion, brain-tumour and abdominal-organ segmentation,
033 one nnU-Net configuration trained with PALETTE improves
034 significantly over all five competing methods, on both Dice
035 and boundary distance.

1. Introduction

036
037 Medical image segmentation models are trained on one ac-
038 quisition setting and are frequently applied to another: a
039 different scanner, sequence, or imaging modality than the
040 one seen during training. MRI contrast is a protocol choice
041 rather than a physical constant, so the same anatomy yields
042 incompatible intensity distributions across sequences, and a
043 network that has only seen T1-weighted brain images can
044 fail outright on T2-weighted or FLAIR images of the same
045 anatomy [5, 23]. Acquiring and annotating training data
046 for every contrast a model might encounter is not practical:
047 manual segmentation is expensive, and clinical sites rarely
048 share raw data across institutions [22, 32]. The single-
049 source domain generalization setting — train on one la-
050 belled dataset in one contrast, generalize to unseen contrasts
051 and modalities without any target-domain data — is there-
052 fore the setting that matters for deployment, and the one a
053 growing body of work addresses [25, 29, 35, 37].

054 The dominant single-source strategy is label-generative
055 domain randomization, popularized by SynthSeg [5]. Given
056 a dense anatomical label map — a voxel-wise parcellation
057 of the whole image, not the sparse structure-of-interest an-
058 notation a segmentation dataset normally provides — in-
059 tensities are resampled from a per-label Gaussian mixture,
060 producing images of arbitrary, unrealistic contrast that force
061 the network to ignore contrast and rely on shape and lay-
062 out [5]. This is effective, but carries two structural costs.
063 First, it is *label-driven*: the dense parcellation it resamples
064 from is a richer annotation than the task itself requires, so
065 generating synthetic contrast costs exactly the expensive la-
066 labelling the method aims to reduce reliance on. Second, and
067 less discussed, it *does not preserve texture*: replacing every
068 labelled region with piecewise Gaussian noise discards the
069 real gradients, partial-volume boundaries, and unlabelled
070 fine structure the acquired scan contains. Layout survives;
071 texture does not.

072 We ask whether discarding texture is necessary for con-
073 trast robustness, or merely the path of least resistance once
074 a dense label map is assumed. Our central finding is that the

answer depends on the *task*: preserving real intensity structure matters where the segmentation target is separated from its surroundings by a genuine anatomical interface, and not where it is not. That texture carries diagnostic signal in exactly the tasks where no such interface exists is established independently of us — texture features discriminate MS lesions from normal-appearing white matter [36], and diffuse glioma infiltrates beyond every margin conventional MRI shows [1, 2] — and our contribution is to show that this distinction governs whether a texture-preserving augmentation helps at all.

We exploit it with **PALETTE**, a single-source augmentation that needs no dense label map. **PALETTE** partitions the foreground of a scan into intensity classes with unsupervised 1-D *k*-means, further subdivides each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap to the *real* intensities within each sub-region. Because the transform remaps existing intensities rather than replacing them with sampled noise, edges, gradients, and partial-volume detail largely survive within each sub-region; only the region-to-region contrast relationship is randomized, including sign inversions no physical acquisition protocol produces. The optional final step is the only one that uses labels, and it uses only the sparse task labels already present in any segmentation dataset — never a dense parcellation.

We test the mechanism claim directly rather than inferring it from aggregate scores. Holding the *k*-means/Voronoi partition and the per-region statistics fixed, we replace only the real-intensity affine remap with label-style noise fill, matching SynthSeg’s resampling inside **PALETTE**’s own geometry. The swap costs 7.7 and 7.2 Dice on the two tasks whose targets have no tissue interface, against +1.1 and −1.1 on the two that do — a separation by target type, not a uniform offset. An independent gradient-orientation measurement confirms that texture is what changes at that exact step.

Empirically, training a single nnU-Net [14] configuration — fixed across datasets, with only the epoch budget varying — we evaluate cross-contrast and cross-modality transfer on four segmentation tasks: healthy brain, MS lesion, brain tumour, and abdominal organ. We compare against a no-augmentation baseline, both variants of SynthSeg (the original dense-label-driven variant, and an EM variant with no public implementation that we reimplement), SRCSM [30], and Auglab [23], a published augmentation pipeline designed for cross-modality training and the strongest competitor. **PALETTE** improves significantly over every competing method on both Dice and boundary distance at the task level. No competitor is reliably second: the runner-up changes identity across settings, so **PALETTE** is the only method at or near the top throughout.

In summary, we make the following contributions:

- **A task-level account of when texture preservation matters.** Preserving real intensity structure helps where the segmentation target has no anatomical interface separating it from its surroundings, and not where it has one. We state this as a prediction and test it with a controlled dissociation rather than asserting it post hoc.
- **A causal ablation isolating texture.** Holding the intensity and spatial partition and the per-region statistics fixed and swapping only the real-intensity remap for label-style noise fill separates the four tasks by target type. An independent gradient-orientation measurement confirms that texture, not the partitioning scheme, is what changes at that step.
- **PALETTE**, the single-source contrast augmentation that exploits this, requiring no dense anatomical label map and using only the sparse task labels a segmentation dataset already provides.
- **Breadth evidence.** Across cross-contrast and cross-modality transfer on healthy-brain, MS-lesion, brain-tumour and abdominal-organ segmentation, with one configuration reused across datasets, **PALETTE** improves significantly over all five competing methods on both Dice and boundary distance.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation (UDA), which assumes unlabelled target images and solves an easier problem [28]. Most medical DG methods are *multi-source*, interpolating across several labelled domains by meta-learning [20], feature alignment [32], or frequency-space exchange [22]. Recent computer-vision entrants follow the same recipe: MADGNet couples a multi-frequency architecture with multi-domain training [24], and foundation models such as SynthFM pursue modality-agnostic segmentation at scale [27], while diffusion-based stylization normalizes appearance but relies on a generative prior and on source diversity [4]. We instead work in the *single-source* regime — one labelled dataset, no target data, no second source — which cannot exploit cross-domain interpolation and makes architecture-coupled methods orthogonal: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [14].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [5]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary con-

trast that force contrast-invariance. This family carries two structural costs. It is *label-driven* — a dense anatomical label map is required per subject, and SIAM shows the dependence is on label *quality*, needing a few high-grade manually curated templates [31]; producing such maps is expensive in expert hours. It is also *texture-destroying*: per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter lesions [17] and MS lesions [6], and sits alongside training-free per-scan generative fitting such as SAMSEG [7]. We take SynthSeg’s original (noEM) variant — which requires a dense anatomical label map — and its EM variant — which does not need a dense map but still uses the sparse task labels naturally available in a segmentation dataset to estimate its per-class intensity model — as baselines; as the EM variant has no public code, a faithful re-implementation is itself a contribution.

2.3. Single-source intensity and appearance augmentation

A second line manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [14, 23] is one such GPU-accelerated framework and is the strongest competitor in our experiments, and BigAug follows the same stacked-transform philosophy [37]. Our method inserts the PALETTE transform as one additional operator inside this pipeline, so the comparison against Auglab directly isolates the contribution of PALETTE. Closer to our operator are random-filter methods: GIN applies randomly re-initialized convolutions per forward pass to produce appearance variants requiring no label map in true 3D with no trained parameters [25], RandConv establishes the principle for natural images [35], SLAug perturbs intensities per class using the task mask [29], and diffusion variants add learned generators [8]; a parallel frequency-domain line perturbs the Fourier amplitude spectrum rather than the image itself [21, 26, 38]. These share our single-source stance — GIN especially is a direct peer — but vary appearance globally, spatially or spectrally, whereas our operator partitions the foreground into intensity classes and Voronoi sub-regions and applies a signed affine remap of the *real* intensities within each. The closest peer is semantic-aware random convolution [30], which also localizes the perturbation region-wise in true 3D, but differs on both axes that define our operator: it delineates regions from the *ground-truth label map*, reintroducing the task-label dependence we avoid, and applies a random convolution within each, which alters local gradients, whereas we partition by unsupervised *k*-means and Voronoi partition and apply an affine remap

that leaves gradients intact. Because the remap is affine and region-local, edges and texture survive while contrast is randomized, preserving exactly the signal label-generative synthesis discards, without any anatomical labels.

2.4. The evaluation gap

We are not aware of prior work reporting our protocol: training on a *single real input modality* and testing on held-out modalities of the *same task and dataset*, across a benchmark spanning brain, abdominal and tumour anatomy in both MRI and CT. Existing single-source evaluations are either cross-dataset but same-modality — e.g. FLAIR MS-lesion models across sites [3] on cohorts such as the public 3D-MR-MS data [19] — or contrast-agnostic-by-synthesis, where training data already span all contrasts and “single-modality” refers only to the test input. The one-real-modality-in, held-out-modality-out axis is the harder setting, and the one we target.

3. Method

We describe PALETTE (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE transform

Let x denote the intensity volume of a training scan, min-max normalised to $[0, 1]$, and $F \subset \Omega$ its foreground voxels (an intensity threshold; $x_v > 0.01$ in our implementation). PALETTE applies four steps at every training iteration, illustrated end-to-end in Fig. 1 on one real CHAOS (abdominal MRI) and one real OnHarmony (brain MRI) volume.

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$ (Fig. 1, column 2). This partition is unsupervised and requires no anatomical label map: it groups voxels by intensity alone, so the same operator applies unmodified to any input image regardless of its modality or contrast — CT, T1, T2, FLAIR, or otherwise. With probability 0.10 this step is skipped and a single global remap is applied instead (step 3 below, with $C = 1$).

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi partition seeded with points sampled within C_i (Fig. 1, column 3 shows the combined effect of this step and the remap of step 3), giving sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different

sub-regions and receive independent contrast perturbations. Each class is subdivided into m_i sub-regions sampled from $\{2, \dots, 10\}$ with probability 0.60, and left as one region (no sub-split) with probability 0.40.

3. Signed affine remap. For each sub-region R , let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v , it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). The result is optionally Gaussian-blurred (a σ sampled from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, weighted toward no blur).

4. Anatomical-label affine remap (optional). Steps 1–3 are unsupervised: the k -means/Voronoi partition groups voxels by intensity and spatial proximity, which need not coincide with the true anatomical boundary between two structures. When two adjacent tissues happen to share similar intensity at a given training iteration, the partition can place both sides of a real label border inside the same region, so step 3 assigns them the same target contrast and the boundary is, by chance, erased rather than randomized. This last step corrects for that: for each real anatomical label ℓ present in the volume, with probability 0.5 (skipped below 4 foreground voxels) we resample a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again*, restricted to ℓ 's voxels only, **on top of the step-3 output** – i.e. this affine remap acts on the already-remapped (step 1–3) intensities, not on the original voxel values, so it composes with rather than replaces the unsupervised remap (Fig. 1, column 4 shows the resulting per-label flat-remap value, computed prior to this step). The voxels of ℓ are therefore *re-assigned* to a new region whose contrast is drawn afresh, but they *inherit* the internal structure the k -means and Voronoi steps already imposed on them: the remap is affine, so relative intensity variation inside ℓ survives. Because the label assignment is independent of whatever k -means/Voronoi region a voxel fell into, this step reintroduces a full-strength contrast discontinuity exactly at the true label border regardless of whether the

unsupervised partition happened to preserve it – sharpening the boundary cue the network can rely on, not the underlying anatomy. The step only ever consumes the segmentation labels the network is already being trained to predict – whatever labels happen to be annotated in the training set, including sparse or partial annotations – so it requires no additional segmentation effort beyond what training already assumes. It is also optional: it can be disabled entirely (`label_remap_prob=0`), in which case PALETTE degrades gracefully to steps 1–3 alone, e.g. on an unlabelled dataset. A final foreground z-score normalisation is applied after an optional second blur pass (same σ distribution as step 3), giving the training input shown in the last column of Fig. 1.

Per-step stochasticity, summarised: step 1 is replaced by a single global remap with probability 0.10; each intensity class is Voronoi-split with probability 0.60 (else left as one region); step 3 is always applied to whatever partition results from steps 1–2; step 4 is applied independently per label with probability 0.5.

3.2. Training protocol

PALETTE is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [14, 23], alongside that pipeline's standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k -range, sub-region count, blur schedule, and α range given in §3.1 – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE differs from label-generative synthesis (e.g. SynthSeg) along two axes at once: the partition it perturbs (unsupervised k -means + Voronoi vs. a supplied anatomical label map as the sole generative substrate) and the perturbation itself (an affine remap of real intensities vs. re-sampling from a per-label Gaussian mixture). Comparing PALETTE to SynthSeg directly therefore cannot attribute an advantage to either axis in isolation.

We construct a controlled variant, *noise-fill*, that holds the k -means/Voronoi partition and the per-region statistics (μ_R, α_R) from Eq. (1) exactly fixed, but fills each sub-region with samples drawn from a Gaussian with that region's $(\mu_R, \alpha_R \sigma_R)$ instead of remapping the real intensities inside it – i.e. the same partitioning scheme, with SynthSeg-style noise substituted for the affine remap. Because the partition and target statistics are identical to PALETTE's own, any difference in downstream performance between PALETTE and noise-fill isolates the one variable that changed: whether real texture is preserved

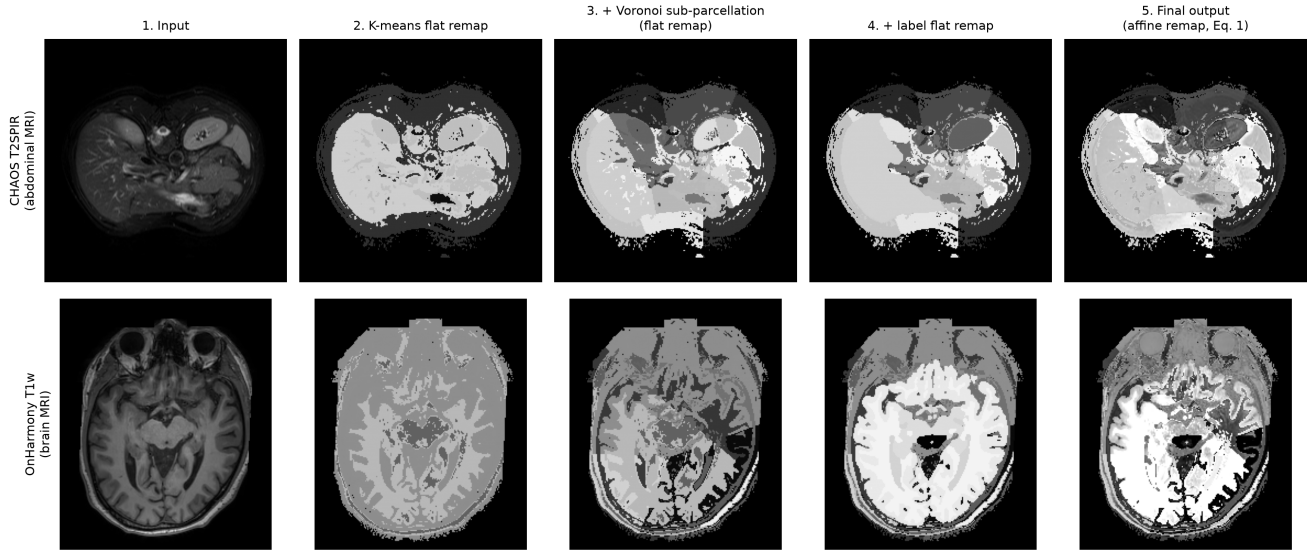


Figure 1. The PALETTE pipeline applied to one real CHAOS T2SPIR volume (top) and one real OnHarmony T1w volume (bottom). For display only, every panel is min-max normalised to $[0, 1]$ (the foreground z-score used at training time is never applied for visualisation); no panel is generated by an external model – every value shown is a deterministic function of the same real volume. Column 1: input. Column 2: *k-means flat remap* – each *k*-means class is painted with the precomputed target mean μ of Eq. (1) for its dominant (largest) Voronoi sub-region, i.e. the flat value that sub-region would take under the transform, not an average of observed pixel intensities. Column 3: the same flat-remap logic applied after Voronoi sub-parcellation, so each sub-region is painted with its own precomputed μ . Column 4: each real anatomical label is additionally painted with its own precomputed μ , overwriting the column-3 value on that label’s footprint. Column 5: final PALETTE training input, obtained by applying the signed affine remap of Eq. (1) with texture (not a flat fill) at both the region level and, where labels are available, the anatomical-label level. Columns 2–4 isolate the flat target value chosen *before* the transform is applied with texture; column 5 shows the texture the transform actually produces once that value is used as its mean.

or discarded within each region. We report this comparison separately for structures with no tissue interface (fine internal gradients, low boundary contrast) and structures bounded by one (identified by a sharp, high-contrast border) in Sec. 4, as the texture-preservation hypothesis predicts a widening gap specifically on the former.

4. Experiments

4.1. Setup

Protocol. We train a single real-intensity source modality per dataset and evaluate on held-out modalities of the *same* segmentation task, using one fixed nnU-Net [14] configuration across all settings. No target-domain data, of any modality, is seen during training. Our full method, denoted **Ours** throughout, is PALETTE applied as one additional contrast-augmentation operator inside the Auglab pipeline [23]; the *Auglab* baseline is the same pipeline with the PALETTE operator removed, so the two differ by exactly the PALETTE transform. Every method is trained with the identical nnU-Net schedule — SGD with Nesterov momentum (0.99), the nnU-Net `polyLR` decay from an initial rate of 10^{-2} , 250 mini-batches per epoch, and the combined Dice-plus-cross-entropy loss — and differs only

in the augmentation applied at the input. The epoch budget is fixed per dataset and shared by every method on that dataset (2500 for Brats-GLI, 2000 for ON-Harmony and Open-MS, 200 for CHAOS), and all results are 3-fold cross-validated, as is standard practice in medical image segmentation and so that every comparison rests on per-case statistics rather than a single split. Full configuration details, and per-dataset descriptions (subject counts, acquisition protocols, and label provenance), are given in Secs. 7 and 8.

Datasets. We evaluate on eight source-modality settings drawn from six datasets, spanning abdominal CT/MRI, brain MRI, and brain tumor MRI. In every setting the model is trained on the single source modality named below. Evaluation is performed exclusively on held-out test cases, never on training or validation subjects, and covers both the other contrasts of that task *and* the training contrast itself; no target modality is seen during training.

- **CHAOS** [16] (abdominal organ segmentation): train on either T1-in-phase or T2-SPIR MRI, and test cross-modality on the modalities *not* trained on — the complementary CHAOS contrasts (T1-in-phase, T1-out-of-phase, T2-SPIR) together with held-out CT from AMOS [15] and SLIVER07 [12].

- **ON-Harmony** [33] (31-region healthy-brain segmentation, cross-contrast): train on either T1w or T2w, and test on all contrasts (the complementary structural contrast plus BOLD, DWI, EPI, and GRE). Reference labels were obtained on T1w with FreeSurfer [10] and registered to the other contrasts.
- **Open-MS** [19] (brain MS-lesion segmentation, cross-contrast): train on either FLAIR or T1w, and test on the held-out contrasts of the same lesion-segmentation task not trained on (FLAIR, T1w, T2w).
- **Brats2024-GLI** [9] (post-treatment glioma segmentation, cross-contrast): train on either T1n or T2w, and test on the held-out contrasts not trained on (T1c, T1n, T2-FLAIR, T2w).

Baselines. SynthSeg comes in two variants, which we refer to throughout by how they obtain their per-class intensity model: *noEM*, the released variant, which reads it off a dense anatomical label map; and *EM*, which estimates it from the image by expectation-maximization and therefore needs no dense parcellation. We compare against five methods. (1) *No-augmentation*, the nnU-Net default training recipe. (2) *SynthSeg-noEM* [5], the original label-generative variant (the implementation released by the authors), which synthesises each training image from a dense anatomical label map alone, ignoring the acquired scan entirely. (3) *SynthSeg-EM*, the variant described in the original work but never released publicly, which we re-implement; unlike SynthSeg-noEM it does not require a dense anatomical label map, but it is not dense-label-free either — it is compatible with and relies on the sparse task labels naturally available in a segmentation dataset, which it uses to estimate its per-class intensity model. (4) *SRCSM*, a semantic random-convolution contrast-synthesis augmentation, included as a second strong synthesis-based competitor that is neither label-generative nor a member of the Auglab family. (5) *Auglab*, the published GPU-accelerated augmentation pipeline of Molinier et al. [23]. Auglab is a prior method in its own right, not an ablation of ours, and we report it among the competitors accordingly; our full method (Sec. 4.1) applies PALETTE *on top of* the Auglab pipeline, so the PALETTE-vs-Auglab comparison also functions as the add-one-component test isolating the contribution of the PALETTE transform.

Statistics. All tests are paired at the level of the individual test case, never per-fold, so significance reflects consistency across cases rather than aggregate noise, and all are Holm-corrected [13] for multiple comparisons. For the task-level comparison of Tab. 1 we use a one-sided sign-flip permutation test on the macro-averaged per-case difference, stratified by task so that each task contributes equally regardless of how many test cases it contributes; this matches the

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	overall	p vs ours
<i>Dice</i> ↑						
Baseline	21.7	34.5	31.1	21.0	27.1	1.5e-38
SynthSeg-noEM	8.6	60.5	66.3	4.3	34.9	2.3e-49
SynthSeg-EM	43.6	85.6	67.4	33.9	57.6	3.9e-14
SRCSM	42.1	85.8	66.6	33.0	56.9	3.1e-15
Auglab	46.7	86.3	67.4	34.3	58.7	0.0003
PALETTE (ours)	46.8	87.3	68.2	38.5	60.2	—
<i>HD95</i> (mm) ↓						
Baseline	47.9	110.9	28.6	35.2	55.7	2.6e-30
SynthSeg-noEM	78.9	83.4	7.5	41.9	52.9	2.9e-57
SynthSeg-EM	17.4	25.4	4.7	24.7	18.1	0.0021
SRCSM	18.3	23.9	5.6	24.6	18.1	5.0e-05
Auglab	15.0	27.9	4.7	26.5	18.5	0.0223
PALETTE (ours)	15.2	24.3	4.2	24.3	17.0	—

Table 1. Task-level results. Each column pools that task’s held-out test contrasts *and* both of its source modalities; **overall** weights the four tasks equally. Per-setting tables (eight source-modality settings, all structures, per-fold) are in the supplementary material. p is a Holm-corrected one-sided sign-flip test on the macro-averaged per-case difference, stratified by task. PALETTE improves significantly over *every* competitor on *both* metrics.

equal-weight-per-task estimand of the *overall* column. Per-setting comparisons use the paired Wilcoxon signed-rank test [34].

4.2. Main comparison

Every synthetic-contrast method vastly outperforms the no-augmentation baseline, confirming that cross-contrast augmentation is indispensable under single-source evaluation. SynthSeg-noEM, which resamples intensities from a dense anatomical label map, is competitive only on the brain-MRI settings where such maps are meaningful and collapses on the sparse-label tumor and lesion tasks; we return to this in Sec. 4.3. The re-implemented, label-using SynthSeg-EM closes most of that gap.

No competitor is reliably second. The identity of the runner-up changes from setting to setting. Across the eight source-modality settings the best competing method is Auglab on most, SynthSeg-EM on ON-Harmony T2w and on boundary distance for CHAOS T2spir, and SRCSM on Open-MS FLAIR and on boundary distance for CHAOS T1in. PALETTE is the only method at or near the top everywhere, and on the task-level aggregate it improves significantly over all five competitors on both Dice and boundary distance (Tab. 1). We regard this consistency, rather than the size of any single margin, as the benchmark claim: a plug-in augmentation is useful precisely when it can be applied without knowing in advance which alternative would have been the right one for the task at hand.

Reading the non-significant cells. Every task-level comparison in Tab. 1 is significant; the non-significant margins appear only in the per-setting tables of the supplementary.

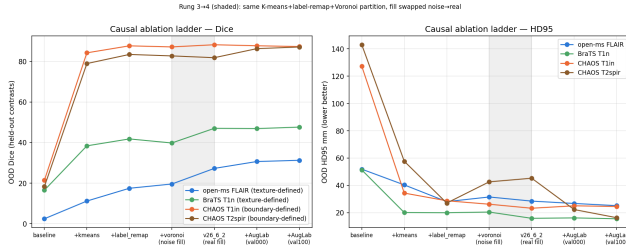


Figure 2. The one-variable test, as a rung sequence. Each step adds exactly one ingredient; the highlighted step swaps label-style noise fill for the real-intensity remap while holding the partition and per-region statistics fixed. Preserving texture is worth +7.7 Dice on the no-tissue-interface lesion task (Open-MS FLAIR) and nothing on the tissue-interface organ task (CHAOS T1in). Tab. 2 reports all four tasks.

Where such a margin is not significant, the appropriate conclusion is equivalence on that setting, not a shortfall. The held-out sets used here are large, so these comparisons are not underpowered; PALETTE attains the best Dice in every setting and the best aggregate on both metrics, though on boundary distance individual settings are occasionally led by a competitor. The advantage is also not a uniform offset – it concentrates sharply on some tasks (Open-MS, 34.3 $\rightarrow$ 38.5) and is near-neutral on others (Brats-GLI, 46.7 $\rightarrow$ 46.8). That pattern is not noise; it is the dissociation we predict and test in Sec. 4.3.

4.3. Causal ablation: the texture dissociation

PALETTE shares its k -means/Voronoi partitioning scheme with a much simpler procedure, so the central question is whether the partitioning alone — not the texture-preserving intensity remap — is what helps. We answer it with an intervention. Along a rung sequence in which each step adds exactly one ingredient (baseline $\rightarrow$ + k -means $\rightarrow$ +label-remap $\rightarrow$ +Voronoi with *noise* fill $\rightarrow$ same partition with *real*-intensity fill $\rightarrow$ +Auglab), the fourth-to-fifth step changes *only the fill*: the partition, the number of sub-regions and the per-region statistics are held fixed, and label-style Gaussian noise is swapped for the real-intensity remap. Any change at that step is attributable to texture preservation alone.

Tab. 2 shows the predicted dissociation in overlap. Preserving real texture is worth +7.2 to +7.7 Dice on the two tasks whose targets are defined by tissue appearance, and nothing at all — +1.07 and -1.14 , i.e. within noise and of inconsistent sign — on the two whose targets are defined by an anatomical interface. The same intervention, applied to four tasks, separates them by target type by a factor of roughly seven. This is the paper’s central result: it identifies *when* texture preservation matters, and it does so by intervention rather than by correlation across methods.

Task	boundary type	Δ Dice	Δ HD95
Open-MS FLAIR	no tissue interface	+7.70	-3.05
Brats-GLI T1n	no tissue interface	+7.22	-4.59
CHAOS T1in	tissue interface	+1.07	-2.97
CHAOS T2spir	tissue interface	-1.14	+4.20

Table 2. The one-variable test. Effect of replacing label-style noise fill with the real-intensity remap, *holding the partition and per-region statistics fixed*, on out-of-domain performance. On *Dice* the effect is large on both no-tissue-interface tasks and absent — in one case mildly negative — on both tissue-interface ones. On *HD95* the separation is not clean: boundary distance also improves on CHAOS T1in, and only CHAOS T2spir degrades. HD95 in mm; negative is better.

We report boundary distance for the same intervention and note that it does *not* separate as cleanly: HD95 improves on three of the four tasks, including the tissue-interface CHAOS T1in (-2.97 mm), and degrades only on CHAOS T2spir. Texture preservation therefore appears to help boundary localisation somewhat more broadly than it helps overlap, and we restrict the dissociation claim to Dice rather than presenting HD95 as confirming it.

Two properties of this test are worth stating explicitly. First, it is not a comparison between methods but between two versions of the same method that differ in one component, so it cannot be explained by differences in augmentation strength, partition granularity or training schedule. Second, the tissue-interface tasks are not tasks on which PALETTE performs poorly — PALETTE remains at or near the top of every setting in Tab. 1. It is specifically the *texture* ingredient that is inert there, while the remaining ingredient, the wider space of region-to-region contrast relationships, continues to help.

Why the margin over Auglab is thin where it is. The same gradient-orientation measurement that isolates texture in Sec. 4.3 also explains the size of the remaining gap. Auglab is *not* a texture-destroying method: measured against the source scan it retains a normalized-gradient-field similarity of 0.772/0.777 (FLAIR/T1w), far above the label-style methods (SynthSeg-EM 0.461/0.422, SynthSeg-noEM 0.332/0.319) and only modestly below PALETTE’s 0.808/0.894 (Sec. 12). Where the target has no tissue interface and texture is therefore the operative cue, two methods that both largely preserve texture should be expected to land close together — which is what we observe. The large margins appear against the methods that discard texture, and the thin ones against the method that does not.

Supporting analyses. A component ablation separating the contributions of the k -means partition, the Voronoi sub-

parcellation and the signed remap is given in Sec. 11. An independent, segmentation-free measurement of texture fidelity – gradient-orientation agreement between each augmented volume and its source – confirms that texture is what changes at the fill-swap step and is reported in Sec. 12. Per-setting tables for all eight source-modality settings are in Sec. 9.

5. Discussion and Limitations

Consistency, not margin, is the benchmark claim. No competing method is reliably second-best. Across the eight source-modality settings the runner-up changes identity – Auglab [23] on most, SynthSeg-EM on on-harmony T2w and on boundary distance for CHAOS T2spir, and SRCSM on open-ms FLAIR and for boundary distance on CHAOS T1in. PALETTE is the only method at or near the top of every setting, and at the task level it improves significantly over all five competitors on both Dice and boundary distance (Tab. 1). Where a per-setting margin is not significant the correct reading is equivalence on that task rather than a shortfall: the test sets here are large, so these comparisons are not underpowered. The claim we make is therefore *equivalent-or-better everywhere, and better on aggregate*, which is a stronger property for a plug-in augmentation than a large win on any single benchmark.

The causal ablation isolates a mechanism, not a full explanation. The noise-fill comparison in Sec. 4.3 shows that texture preservation, specifically, drives the advantage on no-tissue-interface structures; it does not by itself explain the smaller, more uniform gains observed on tissue-interface structures, which likely stem from the wider space of region-to-region contrast relationships (including the sign inversions in Eq. (1)) that no single real acquisition protocol supplies. Disentangling these two effects further is left to future work.

SynthSeg-noEM’s collapse on sparse-label tasks is expected. The large gap between SynthSeg-noEM and the other methods on Brats2024-GLI and Open-MS (Tab. 1) is not an implementation artifact but a direct consequence of what SynthSeg-noEM requires. It generates images by resampling intensities from a *dense* anatomical label map covering the whole volume; on sparse-label segmentation tasks — a single glioma or a handful of MS lesions floating in an otherwise unlabelled brain — there is no such dense map, so the generated images carry almost no anatomical structure and the network cannot learn. SynthSeg-noEM is therefore expected to be competitive only where dense label maps are meaningful (the brain-MRI settings) and to collapse elsewhere; this is a property of the method’s input requirements, not a caveat about our re-implementation.

Why the dissociation follows from the anatomy. Preserving real intensity structure matters where the target is defined by tissue *appearance*, and not where it is defined by an anatomical *interface*. Liver, spleen and kidney are encapsulated organs bounded by fat planes – a genuine physical interface, which is why edge-following methods such as geodesic active contours have long succeeded on them [16, 18]; an augmentation that scrambles internal texture while preserving that interface loses little. Diffuse glioma has no such interface: tumour cells infiltrate beyond both the contrast-enhancing margin and the peritumoral FLAIR abnormality [1, 2], so the target must be inferred from appearance rather than followed along an edge. MS lesions are likewise defined by tissue signal characteristics rather than an anatomical border [36]. This account is an interpretation of the ablation, not an independent proof of it; the causal evidence is the intervention in Sec. 4.3.

The label step needs only the annotation training already assumes. PALETTE’s one label-consuming step uses whatever segmentation labels the training set happens to carry – sparse, partial, or task-specific – and never a dense anatomical parcellation. This is the practical difference from label-generative synthesis: SynthSeg’s dense requirement means the augmentation costs an annotation richer than the task, whereas here the step is free at training time and can be switched off entirely on an unlabelled dataset, in which case PALETTE degrades gracefully to its unsupervised steps.

Scope. PALETTE targets the single-source setting: one labelled training modality, no access to target-domain data of any kind. It is input-level and architecture-agnostic, and we do not claim it subsumes multi-source domain generalization methods that have access to several labelled source domains; the two are complementary rather than competing solutions to the same problem.

6. Conclusion

We asked when preserving real image texture matters for contrast-robust segmentation, and found that the answer is a property of the task rather than of the augmentation: it matters where the target is defined by tissue appearance, and not where it is defined by an anatomical interface. A controlled intervention – holding the partition and per-region statistics fixed and swapping only label-style noise fill for a real-intensity remap – separates four tasks cleanly along exactly that line, worth +7.7 and +7.2 Dice on the two no-tissue-interface tasks and nothing (+1.1, −1.1) on the two tissue-interface tasks. PALETTE, the single-source augmentation built on this observation, partitions the foreground by unsupervised k -means and a Voronoi sub-parcellation and ap-

plies a signed affine remap of the real intensities within each sub-region, requiring neither a dense anatomical label map nor any target-domain data. Across eight cross-contrast and cross-modality settings spanning abdominal, brain and brain-tumor imaging, a single nnU-Net configuration trained with PALETTE attains the best Dice in every setting and improves significantly over all five competing methods on both Dice and boundary distance. We regard the task-level account as the more durable contribution: it predicts in advance where a texture-preserving augmentation will and will not pay off, and that prediction is testable on any new task.

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PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

875 7. Training configuration

876 All methods share one nnU-Net [14] configuration; only
877 the input augmentation differs. Nothing below is tuned per
878 dataset.

879 **Training schedule.** 3D full-resolution nnU-Net, SGD
880 with Nesterov momentum 0.99, initial learning rate 10^{-2}
881 with the nnU-Net `polyLR` decay, 250 mini-batches per
882 epoch, combined Dice and cross-entropy loss, deep super-
883 vision on. Patch size, spacing and batch size are whatever
884 the nnU-Net planner selects for that dataset, identically for
885 every method. The epoch budget is fixed per dataset and
886 shared by all methods on it: 2500 (Brats-GLI), 2000 (ON-
887 Harmony), 2000 (Open-MS), 200 (CHAOS). Every number
888 reported in this paper is a 3-fold cross-validated average;
889 folds are the same across methods.

890 **PALETTE hyper-parameters.** Set once and never tuned
891 per dataset: k -means class count drawn per sample from
892 $\{2, \dots, 6\}$; Voronoi sub-parcellation of each intensity class;
893 signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied per sub-
894 region as in Eq. (1); per-label affine decoupling applied to
895 50% of labels. The transform is applied on the GPU, on-
896 the-fly, to a fixed 50% of training iterations (`train050`); the
897 remaining half of iterations use the un-transformed real im-
898 age.

899 **Validation-time configuration.** `val000` (the configura-
900 tion reported in the main paper) applies no synthetic con-
901 trast to the validation split, matching every competing
902 method and keeping model selection like-for-like. `val100`
903 applies PALETTE to the entire validation split; it is reported
904 in Sec. 10.

905 **Baseline configurations.** SynthSeg-noEM uses the au-
906 thors' released generator with its default priors. SynthSeg-
907 EM is our re-implementation of the variant described but
908 not released in the original work; it estimates a per-class
909 intensity model from the sparse task labels available in
910 each dataset via expectation-maximization, and is otherwise
911 identical in its sampling procedure. SRCSM and Auglab
912 use their published default configurations. No competing
913 method was re-tuned in our favour, and all use the same
914 folds, schedule and epoch budget as above.

Compute. Training was performed on NVIDIA L40S and
H100 GPUs. A single fold costs from under an hour
(CHAOS, 200 epochs) to roughly three days (Brats-GLI,
2500 epochs); the full study is dominated by the 7 methods
 $\times 8$ settings $\times 3$ folds main comparison plus the ablation
ladders of Sec. 4.3.

921 8. Datasets and evaluation protocol

922 All evaluation is performed on held-out test cases only. For
923 each task the model is trained on a single source contrast
924 and evaluated on every contrast of that task's test split, in-
925 cluding the training contrast itself; training and validation
926 subjects never appear in any reported number.

ON-Harmony [33] — 31-region healthy-brain segmenta-
tion. Reference labels are *silver standard*: they were pro-
duced automatically with FreeSurfer [10] on the T1w vol-
ume of each subject and then registered to that subject's
other contrasts, which are acquired in the same session.
They are therefore not manual annotations, and absolute
Dice on this task should be read as agreement with an auto-
mated reference rather than with a human rater; the compar-
ison between methods, which all use the identical reference,
is unaffected.

Open-MS [19] — MS-lesion segmentation with manual
consensus annotations, evaluated across FLAIR, T1w and
T2w.

Brats2024-GLI [9] — post-treatment glioma sub-region
segmentation, evaluated across T1n, T1c, T2w and T2-
FLAIR.

CHAOS [16] — abdominal organ segmentation (liver,
spleen, both kidneys), evaluated across the CHAOS MR
contrasts and, as a cross-modality transfer, on held-out CT
from AMOS [15] and SLIVER07 [12].

Per-dataset subject counts, acquisition parameters and
label definitions will be tabulated here in the camera-ready
version.

950 9. Per-setting results

951 Tab. 1 in the main paper reports results at the task level,
952 pooling each task's held-out test contrasts and both of its
953 source modalities. This section reports the same evalua-
954 tion broken out into the eight individual source-modality

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	25.3	8.7	43.1	40.3	46.3	46.4
Brats-GLI T2w	18.1	8.4	44.1	43.8	47.1	47.3
CHAOS T1in	36.1	64.4	85.7	86.5	86.7	87.4
CHAOS T2spir	32.8	56.6	85.4	85.1	86.0	87.2
ON-Harmony T1w	27.5	66.0	67.3	66.4	67.6	68.6
ON-Harmony T2w	34.8	66.7	67.5	66.7	67.2	67.7
Open-MS FLAIR	25.2	4.6	32.8	34.6	32.8	39.5
Open-MS T1w	16.7	4.0	35.1	31.4	35.8	37.4

Table 3. Per-setting all-class Dice (%), averaged over folds and over every held-out test contrast of that setting. *Ours* = PALETTE (train050/val000), the configuration used throughout the main paper. ↑

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	41.0	78.0	18.4	20.8	15.1	15.4
Brats-GLI T2w	54.9	79.7	16.5	15.7	14.8	15.0
CHAOS T1in	102.2	76.6	28.1	23.1	31.9	24.3
CHAOS T2spir	119.7	90.1	22.7	24.8	23.9	24.3
ON-Harmony T1w	33.4	10.4	5.4	6.7	5.2	4.3
ON-Harmony T2w	23.7	4.6	4.0	4.6	4.2	4.0
Open-MS FLAIR	39.9	41.5	23.6	23.8	26.6	24.3
Open-MS T1w	30.5	42.3	25.9	25.5	26.3	24.2

Table 4. Per-setting HD95 (mm), same protocol as Tab. 3. ↓

settings. Every number is a cross-fold, cross-class average over all held-out contrasts of that setting, under the protocol of Sec. 4.1.

Two observations are easier to see here than at task level. First, PALETTE attains the best Dice in *all eight* settings. Second, the identity of the runner-up is not stable: Auglab is the strongest competitor on six of the eight Dice settings, SRCSM on Open-MS FLAIR, and SynthSeg-EM on ON-Harmony T2w; on HD95 the runner-up changes again. No competing method can be relied upon as a default, which is the practical argument for a plug-in transform that is at or near the top everywhere.

10. Synthetic contrast at validation time

Throughout the main paper PALETTE is applied with synthetic contrast during training only (*train050/val000*), because every competing method is likewise evaluated with a real-contrast validation set; this keeps the comparison exactly like-for-like. PALETTE can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*train050/val100*).

Validation-time randomization improves the aggregate on both metrics (Tab. 5), most visibly on boundary distance (17.0 → 16.2 mm overall, and 24.3 → 21.7 mm on CHAOS). The effect is not significant at task level, and Open-MS moves slightly the other way, so we do not claim

Task	Dice ↑		HD95 (mm) ↓	
	val000	val100	val000	val100
Brats-GLI	46.8	47.3	15.2	14.7
CHAOS	87.3	87.5	24.3	21.7
ON-Harmony	68.2	68.4	4.2	4.1
Open-MS	38.5	37.9	24.3	24.2
overall	60.2	60.3	17.0	16.2

Table 5. Applying PALETTE at validation time as well as at training time. The two configurations are statistically indistinguishable at task level (Dice $p = 0.66$, HD95 $p = 0.99$), so we report the like-for-like *val000* configuration in the main paper and note *val100* as a free additional gain – it requires no extra data, annotation or training cost, only applying the same transform to the validation split.

it as a separate contribution; we report it because it costs nothing and because it indicates that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

11. Component ablation

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in general, we build an additive ablation ladder on top of the no-augmentation anchor (*Base*, Sec. 4.2): first the unsupervised k -means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE in its entirety and is denoted **Ours** in this ladder — identical to the standalone (no-Auglab) configuration used throughout Sec. 4.2. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE’s own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k -means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

Composability with Auglab. PALETTE is deployed, throughout this paper, as one additional operator inside the Auglab pipeline rather than as a replacement for it (Sec. 4.1); Auglab is a full prior augmentation method in its own right, not a component of PALETTE, so stacking it on top of the ladder above is a composability/deployment check, not a further mechanism-ablation step. Tab. 6 keeps the two separate: the top five rows isolate PALETTE’s internal mechanism, evaluated with no Auglab throughout,

Component	CHAOS	Open-MS	BraTS	ON-Harmony
Base (no augmentation)	36.1	25.2	23.8	34.8
+ k -means partition	84.8	30.6	—	—
+ label remap	87.7	34.6	—	—
+ Voronoi sub-parcellation	87.2	36.1	—	—
+ affine remap (Ours)	87.9	40.4	—	—
+ Auglab (deployed)	87.3	39.5	44.4	67.7

Table 6. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE component on top of the row above; all-modality Dice (%), averaged over held-out test contrasts, folds 0–2, train fraction 50%. Intermediate k -means/label-remap/Voronoi rungs and the standalone **Ours** rung are available for Open-MS and CHAOS; BraTS and ON-Harmony currently report only Base and the deployed configuration, with the intermediate rungs and standalone **Ours** pending.

Method	FLAIR	T1w
PALETTE	0.808	0.894
Auglab	0.772	0.777
SynthSeg-EM	0.461	0.422
noise-fill	0.453	0.401
SynthSeg-noEM	0.332	0.319

Table 7. Texture fidelity measured by NGF similarity to the source scan (mean over $n=30$ Open-MS scans; higher = more source texture retained, $\approx 1/3$ = isotropic-noise floor). PALETTE preserves the most texture; the noise-based methods sit near the floor.

$p = 1.2 \times 10^{-6}$), while the label-style methods (SynthSeg-EM, noise-fill) and especially SynthSeg-noEM sit near the noise floor — exactly the ordering the mechanism predicts, and consistent with the downstream Dice gaps of Tab. 2.

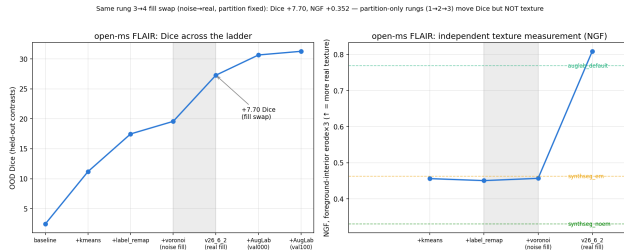


Figure 3. Downstream Dice (left) against an independent, segmentation-free measurement of texture fidelity (right, NGF gradient-orientation agreement with the source volume) across the rung sequence. The three noise-fill rungs sit flat at the texture floor while their Dice still rises – those gains come from richer contrast randomization, not texture. Both quantities jump together at the fill swap, which is the step the texture hypothesis predicts.

while the bottom row reports PALETTE plus Auglab — the configuration used everywhere else in this paper — set apart by a rule rather than presented as the next rung of the same ladder.

12. Direct measurement of texture fidelity

The Dice ablation above The ablation of Sec. 4.3 shows that texture preservation *matters* downstream; here we measure the texture itself, directly and without reference to any segmentation. For each method we compute the normalized-gradient-field (NGF) similarity [11] between the augmented image and the source scan — an alignment-of-gradients measure that is high when local edge and texture structure is retained and near the isotropic-noise floor ($\approx 1/3$) when it is destroyed. Tab. 7 reports mean NGF over $n=30$ scans on the two Open-MS contrasts. PALETTE retains far more of the source texture than any noise-based method and significantly more than Auglab (FLAIR $p = 7.6 \times 10^{-3}$, T1w